Willow WU

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New-Grad Software Engineer — Full-Stack / Backend · TypeScript/React · Python/SQL · WebSockets / Distributed Systems

EDUCATION

Columbia University *New York, NY*
M.S. in Computer Science *(Machine Learning Track)* — GPA 3.96/4.0 *Sep 2025 – Dec 2026*
Relevant Coursework: Analysis of Algorithms, Applied Machine Learning, Deep Learning, Natural Language Processing, Engineering Software-as-a-Service (SaaS), Advanced Spoken Language Processing

Hong Kong Baptist University (HKBU) *Hong Kong*
B.Sc. in Data Science *Sep 2021 – Jun 2025*

EXPERIENCE

Microsoft — Software Engineer Intern *Summer 2026*
OfficePLUS AI Assistant

- Contributed to refactoring a multi-session, token-streaming LLM chat engine (React/TypeScript, WebSocket RPC), consolidating **8** run-tracking tables into a 2-table session model and eliminating active-run eviction, a state-corruption failure mode implicated in ∼**40%** of user downvotes.
- Built a routing-verification probe proving **500/500** concurrent runs routed correctly with **zero** cross-session cross-talk; audited 6 error-fallback branches (2 fired live) to prevent regressions.
- Shipped a change through the full release chain (PR → rebase → Azure DevOps pipeline → signed artifact → handoff); fixed dual-instance concurrency and input-loss bugs.

Columbia University — Teaching Assistant, Introduction to Databases *New York, NY*
Prof. Kenneth Ross *Sep 2025 – Present*

- Supported **100+** students in relational schema design, SQL implementation, and query-plan analysis; graded exams and debugged indexing/performance issues.

PROJECTS

FlowTrack — Full-Stack Job-Search Tracker *Columbia ESaaS* *2025*

- Built a full-stack tracker in **Ruby on Rails** + **PostgreSQL** with a live **Sankey diagram** for Applied → Interview → Offer; **RSpec** + **Cucumber** tests; deployed to Heroku.

Evidence of Life — AI Life-Tracking Web App *Startup Studio* *2026*

- Built a solo full-stack React/TypeScript SPA on **Supabase** (Auth, Postgres, Edge Functions) with AI-assisted input classification and map/calendar views; **51** passing Vitest tests.

Oracle OS — AI Symbolic-Reflection Product *Hackathon* *2026*

- Architected a full-stack app (**React 19** + Express/TypeScript) with deterministic engines and zod-validated APIs; isolated LLM calls to reflect/chat endpoints only.

PUBLICATIONS

You Wu, Junwei Su, Sirui Zhang, Zhijian Li. Explainable Dynamic Graph Neural Networks for Predictive Maintenance in Vehicle Chassis Systems. **ACML 2025** (PMLR).

SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Ruby, C++
Web / Backend: React, Node.js, Ruby on Rails, Django, REST APIs, WebSockets, Supabase, Tailwind, shadcn/ui
Foundations: Data Structures & Algorithms, Object-Oriented Programming (OOP), Distributed Systems, Software Development Lifecycle (SDLC), Agile
Infra / Testing: Azure DevOps, CI/CD, Heroku, Vercel; RSpec, Cucumber, Vitest
Databases: PostgreSQL, SQLite, DuckDB

HONORS & AWARDS

Kaggle Silver Medal (Top 4.2%, 2024) • **MCM Honorable Mention** (Top 20%, 2024) • **CUMCM Provincial 2nd Prize** (Top 16%, 2023) • Merit Scholarship (Top 15%)